SE423 Software Project Management

Artifact 3: Estimation and Scheduling

Scenario 11: National Employment and Skill Matching Platform

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1. Work Breakdown Structure (Hybrid WBS)

1.1 Project Planning

This phase establishes the project foundation by defining its objectives, scope, deliverables, team responsibilities, schedule, and required resources. It also identifies key stakeholders, assesses initial project risks, and establishes the procedures for monitoring progress throughout development.

1.2 Requirement Analysis

This stage involves identifying and capturing all the requirements for Fursah. It includes Job Seeker Portal, Employer Portal, Smart Matching Engine, career advisory, training recommendation, analytics dashboard, and professional networking. It also includes the identification of security, bi-lingual requirements, government connectivity, and hosting requirements. These identified requirements are validated by the stakeholders before system design.

1.3 System Design

This is the phase where the approved requirements are converted into a technical design for the platform. It includes the definition of architecture, data structure, user interfaces, Smart Matching Engine, workflow for Natural Language Processing (NLP) processing, security features, and interfaces for approved government services.

1.4 System Development

This phase implements the platform according to the approved system design. This includes development of the Job Seeker Portal and Employer Portal, Smart Matching Engine and the other functions like career guidance, training suggestions, analytics and professional networking. It will also include implementation of Arabic and English language support.

1.5 System Testing

In this stage, it is checked whether the designed platform satisfies the requirements

specified for it. The testing will include testing individual modules, interface testing, testing search and recruiting processes, results matching, security, performance, usability, and multilingual support. Detected errors are fixed, and after that, regression testing is performed to ensure that the fixing process does not spoil any other functionality.

1.6 System Integration and Delivery

In this stage, all the system modules that have been built are integrated and it is ensured that all the modules work in an integrated manner. This involves integration of portals, matching engine, database, and other modules along with connection to authorized government service providers where applicable. The deployment environment is readied and the platform is deployed.

1.7 Scope Review

At this stage, it is confirmed whether the product meets the specified scope of the project and acceptance criteria. The team goes through the deliverables that were delivered, checks for any scope discrepancies and open issues, and ensures that the excluded functionalities were not added to the system by mistake.

2. Release 1: Product Backlog

2.1 Product Backlog Overview

To keep track of all user stories required for building Fursah, we must create a product backlog. We also want to ensure our stories align with the scope from Artifact 1. Moreover, we want all our features covered. The structure of our user stories is as follows: As a [user], I want [feature], so that [benefit]. We will then group the stories into epics, estimate them using story points, and prioritize them based on importance and order.

2.2 Story Point Estimation

We will use the Fibonacci scale to estimate our user stories. The story points help in estimating the complexity, uncertainty, and effort.

Story Points	Meaning
1	Very small and simple change
2	Small story with clear requirements
3	Medium story that needs some design and testing
5	Large story with several parts or connections to other features
8	Very large or complex story with some uncertainty

Priority levels:

- High: Required for the main job-search and hiring process
- Medium: Improve overall user experience, but platform can work without it initially
- Low: Useful additions that we can postpone to after the main features are delivered

2.3 Prioritized Product Backlog

The following table shows our product backlog by the sprints and the priorities within them.

ID	Epic	User Story	SP	Priority	Sprint
US-01	Shared Platform Services	As a user, I want to register as a job seeker or an employer, so that I can access Fursah's features.	3	High	1
US-02	Shared Platform Services	As a registered user, I want to log in and out securely, so that I can access my account safely.	3	High	1
US-03	Job Seeker Portal	As a job seeker, I want to create and maintain a profile with my skills, qualifications, and experience, so that the platform can find opportunities that suit me.	5	High	1
US-04	Employer Portal	As an employer, I want to create and maintain an employer profile, so that job seekers can learn about my	5	High	1

		organization.			
US-05	Shared Platform Services	As a platform administrator, I want job seekers, employers, and administrators to have different access permissions, so that each user only uses the features meant for them.	3	High	1
US-06	Shared Platform Services	As the Ministry (HRSD), I want platform data and infrastructure hosted within Saudi Arabia, so that the project meets its hosting requirement	3	High	1
US-07	Employer Portal	As an employer, I want to create job postings, so that job seekers can find and apply for them.	5	High	2
US-08	Employer Portal	As an employer, I want to specify the skills, qualifications, and other requirements for each opening, so that Fursah can match the job with suitable candidates.	3	High	2
US-09	Job Seeker Portal	As a job seeker, I want to find and browse available employment opportunities, so that I can discover jobs that interest me.	5	High	2
US-12	Job Seeker Portal	As a job seeker, I want to upload my resume and other professional details, so that employers and the matching engine can review my full background.	3	High	2
US-10	Employer Portal	As an employer, I want to edit or remove my job postings, so that my listings stay accurate and up to date.	3	Medium	2
US-11	Job Seeker Portal	As a job seeker, I want to filter jobs by field, location, job type, or experience level, so that I can quickly find relevant results.	3	Medium	2
US-13	Job Seeker Portal	As a job seeker, I want to apply for jobs, so that employers can consider me for the position.	5	High	3

US-14	Job Seeker Portal	As a job seeker, I want to track the status of my applications, so that I know the progress of each one.	3	High	3
US-15	Employer Portal	As an employer, I want to receive and view submitted applications, so that I can review the candidates for my jobs	3	High	3
US-16	Employer Portal	As an employer, I want to view candidate profiles and resumes, so that I can evaluate whether they fit the position.	3	High	3
US-17	Employer Portal	As an employer, I want to update application statuses and track the hiring process, so that applicants know the result of their application.	3	High	3
US-18	Employer Portal	As an employer, I want to schedule interviews with selected candidates, so that the hiring process moves forward.	5	High	3
SP-01	Smart Matching Engine	Spike: Investigate NLP approaches for extracting skills from Arabic and English resumes and job descriptions, and how to calculate a meaningful match score. Timeboxed to 2–3 working days	-	High	4
US-19	Smart Matching Engine	As a job seeker, I want my resume, qualifications, skills, and experience to be analyzed, so that the platform understands my profile	8	High	4
US-20	Smart Matching Engine	As an employer, I want my job descriptions and requirements to be analyzed using NLP, so that they can be compared accurately with candidate profiles.	8	High	4
US-21	Smart Matching Engine	As a user, I want candidate profiles to be compared with job requirements, so that suitable matches are identified.	8	High	4
US-22	Smart	As a job seeker, I want to see job	5	High	5

	Matching Engine	opportunities recommended from my qualifications and skills, so that I can find suitable jobs faster			
US-23	Smart Matching Engine	As a job seeker, I want to see a match percentage for each job, so that I understand how well I fit the position	3	High	5
US-24	Smart Matching Engine	As an employer, I want suitable candidates to be suggested for my openings, so that I can find the right people faster.	5	High	5
US-25	Employer Portal	As an employer, I want to screen applicants based on the job requirements, so that I can shortlist candidates efficiently.	5	High	5
US-26	Shared Platform Services	As a user, I want to use the platform in Arabic or English, so that I can work in my preferred language.	5	High	5
US-27	Shared Platform Services	As a user, I want my personal and employment data, including my resume, to be stored securely and handled according to Saudi data-protection requirements, so that my data is protected from unauthorized access.	5	High	6
US-28	Shared Platform Services	As a platform administrator, I want to manage user accounts, so that the platform stays organized and safe	3	Medium	6
US-29	Shared Platform Services	As a platform administrator, I want to review job postings and remove false or inappropriate content, so that the platform stays trustworthy.	3	Medium	6
US-30	Shared Platform Services	As a platform administrator, I want important actions to be logged, so that I can track suspicious or unusual activity.	2	Medium	6
SP-02	Career Guidance	Spike: Investigate where external training programs and courses can be	-	Medium	7

	and Training	sourced from and how they can be linked to skills. Time-boxed to 2–3 working days.			
US-31	Career Guidance and Training	As a job seeker, I want my skills and interests evaluated to receive tailored career advice, so that I can choose suitable career paths.	5	Medium	7
US-32	Career Guidance and Training	As a job seeker, I want to see the gap between my skills and the skills required for a career path, so that I know what I need to improve.	8	Medium	7
US-33	Career Guidance and Training	As a job seeker, I want training programs and courses recommended for my skill gaps, so that I can become more qualified.	5	Medium	7
US-34	Career Guidance and Training	As a job seeker, I want to see how a suggested training program would help close my skill gap, so that I can choose the most useful one.	5	Medium	7
US-35	Analytics Dashboard	As a Ministry (HRSD) user, I want to view labor-market trends based on platform data, so that I can monitor the job market.	8	Medium	8
US-36	Analytics Dashboard	As a Ministry (HRSD) user, I want to identify skill gaps between the workforce and employer needs, so that I can support workforce planning.	8	Medium	8
US-37	Analytics Dashboard	As a Ministry (HRSD) user, I want data and reports for labor and education policy, so that decisions are based on evidence.	5	Medium	8
US-38	Analytics Dashboard	As a platform administrator, I want the dashboard to be available only to authorized Ministry users, so that sensitive labor-market data is protected.	3	Medium	8
US-39	Professional	As a job seeker or professional, I want	5	Low	9

	Networking	to connect with other users, so that I can build my professional network.			
US-40	Professional Networking	As a user, I want to share career insights, so that others can learn from my experience.	5	Low	9
US-41	Professional Networking	As a user, I want to join industry forums, so that I can discuss professional topics with others.	8	Low	9
US-42	Professional Networking	As a platform administrator, I want to moderate forum content, so that discussions stay professional.	3	Low	9
SP-03	Government Service Integration	Spike: Investigate the interfaces, authorization, and data formats offered by the participating government agencies. Time-boxed to 2–3 working days	-	Medium	10
US-43	Government Service Integration	As the Ministry (HRSD), I want the platform integrated with relevant government services, so that users benefit from existing national services.	8	Medium	10
US-44	Government Service Integration	As a government agency, I want to exchange required information with the platform through approved integrations, so that data stays consistent and authorized.	8	Medium	10
US-45	Government Service Integration	As a job seeker, I want the platform to use information made available through authorized government integrations, so that I enter less information manually.	5	Medium	10
US-46	National Rollout	As an employer from any industry, I want the platform to cover a wider range of industries, so that I can find candidates in my field.	5	Medium	11
US-47	National Rollout	As a user anywhere in Saudi Arabia, I want the platform to remain fast and reliable under nationwide use, so that I	8	Medium	11

		can use it whenever I need it.			
US-48	National Rollout	As the Ministry (HRSD), I want public awareness activities and onboarding material, so that more people adopt the platform.	3	Low	11
US-49	National Rollout	As the Ministry (HRSD), I want the platform deployed for nationwide use after final testing, so that all users across Saudi Arabia can access it.	5	Medium	11

2.4 Backlog Summary by Sprint

Sprint	Focus	Number of items	Story Points
1	User accounts, profiles, access permissions, Saudi hosting	6	22
2	Job postings, job search, resume upload	6	22
3	Job applications, employer review, interview scheduling	6	22
4	Matching spike, resume and job description analysis, matching	4 (3 stories + 1 spike)	24
5	Job and candidate recommendations, match percentage, screening, Arabic/English	5	23
6	Data security, administration, action logging, core platform testing	4	13
7	Career guidance and training recommendations	5 (4 stories +1 spike)	23
8	Analytics Dashboard for the Ministry	4	24
9	Professional networking	4	21
10	Government service integration	4 (3 stories + 1 spike)	21
11	National rollout	4	21
Total		**52**	**236**

Approximately, our team's capacity is about 20-25 story points per Sprint (that lasts 2 weeks). Coming in with the 13 story points, sprint 6 has the lowest load. This isn't a coincidence, this is actually to leave some time for any of the following tasks: integration, system and security testing, defect fixing, and stakeholder feedback. We also must make sure that we can leave room before the core platform is put out.

3. Scrum Technical Process and Schedule

3.1 Sprint Duration and Overall Timeline

Fursah will be carried out using Scrum methodology. The platform's development process will be done with 2 week sprints. Each sprint is responsible for delivering a specific set of features from the Product Backlog. At the end of every sprint the employers, job seekers, and HRSD are able to review the work and provide the team feedback which is used for the next sprint. This enables the team to find problems earlier on and change the platform as requirements change.

Fursah will implement the two- week sprints and which is sufficient enough for the team to finish essential features, such as helping job seekers to apply for jobs and the employers to review the applications, while being short enough for users and HRSD to frequently evaluate and review the work and propose changes early. Before beginning the development, the team will conduct a Sprint 0 which will focus on setting up the development environment and identifying the initial requirements and design.

Release 1 of Fursah will approximately take 14 weeks, with each sprint lasting 2 weeks. Sprint 0 focuses on preparation, followed by six development sprints.

Sprint	Duration	Main Activities	Expected Deliverables
Sprint 0 (Preparation)	Weeks 1-2 (2 weeks)	Preparation for the development environment, set up the initial Product Backlog, define the initial requirements and design.	Set up the development environment and initial Product Backlog.

Sprint 1 (Platform Foundation)	Weeks 3-4 (2 weeks)	Create user registration, login, role based access control, set up job seeker and employer profiles, and certify hosting within Saudi Arabia.	Functional user accounts and profiles.
Sprint 2 (Job Postings and Search)	Weeks 5- 6 (2 weeks)	Enable employers to add, edit, and delete job postings with their requirements and enable job seekers to search and filter available jobs and upload resumes.	Functional job postings and job search.
Sprint 3 (Applications and Hiring)	Weeks 7-8 (2 weeks)	Allowing job seekers to submit their job applications and monitor their progress while enabling employers to review applications, change their status, and schedule for interviews.	Functional job application system.
Sprint 4 (Resume and Job Analysis)	Weeks 9-10 (2 weeks)	Perform the NLP spike, and then develop features that analyze resume and job descriptions and assess them to find matches.	The platform can initially identify skills and matches.
Sprint 5 (Recommend ations)	Weeks 11-12 (2 weeks)	Display matching scores, suggest jobs to job seekers, recommend applicants to employers, and enable support for Arabic and English.	Initial Smart Matching Engine and recommendations for job seekers and employers in both Arabic and English.
Sprint 6 (Security, Testing, and Final Review)	Weeks 13-14 (2 weeks)	Implement secure data storage, administrator tools, and activity logging, after testing all Release 1 features all together, detect and fix any problems and present the completed release to HRSD.	Release 1 is tested and gains approval from HRSD.

Releases 2 and 3 are initially planned at a high level as Sprints 7- 11 in the Product Backlog and will be planned in more detail as their start dates approach, taking into consideration feedback from Release 1.

3.2 Scrum Meetings and Frequency

The table showcases the Scrum meetings that the team will hold for each sprint

Meeting	Frequency	Duration	Purpose
Sprint Planning	At the start of each sprint	Up to 4 hours	Identify the sprint goal and select the user stories the team will work on.
Daily Scrum	Each working day	15 minutes	Update the team on progress, plan and prepare for the day, and identify any problems.
Sprint Review	At the end of each sprint	Up to 2 hours	Provide the finished features to employers, job seekers, and HRSD to gather their feedback.
Sprint Retrospective	At the end of each sprint, after the sprint review	Up to 1.5 hours	Evaluate what went right and discuss the improvements for the upcoming sprint.
Backlog Refinement	Once during each sprint	1-2 hours	Review, split, and estimate the user stories for the next sprints.

3.3 Additional Scrum Elements

The table showcases the additional Scrum elements that will be used in this project

Element	Description	The use in Fursah
Sprint 0	Before development begins the team conducts a preparation Sprint 0.	Focusing on preparation of tools, establishing the initial Product Backlog, and identifying the initial requirements and design prior to Sprint 1.
Epics	Big features are divided into smaller user stories.	Release 1 epics: Shared Platform Services, Job Seeker Portal, Smart Matching Engine, and Employer Portal
Spikes	Brief research with a fixed duration is utilized when the team is uncertain about how to build and implement something.	NLP-based Smart Matching Engine spike (Sprint 4): investigate how to analyze Arabic and English job descriptions, resumes, and identify matching skills since it’s the platform's main technical challenge. 2 additional spikes are scheduled for later releases: Career Guidance and Training (Sprint 7) and Government Service Integration

		(Sprint 10).
Definition of Done (DoD)	A list of conditions each user story must fulfill before it is considered as finished.	The code is reviewed by another developer, passes all the required tests, complies with the acceptance criteria, meets the Saudi Personal Data Protection Law (PDPL), and approval by the Product Owner.

4. Velocity & Estimation

4.1 Estimation Technique (Planning Poker, Fibonacci Scale)

The Fursah team will estimate user stories using Planning Poker with the modified Fibonacci scale already established in the product backlog: 1, 2, 3, 5, and 8 story points. During backlog refinement or Sprint Planning, the Product owner explains the story and its expected value. Team members independently select an estimate and reveal their estimates together. When estimates differ, members explain their reasoning, discuss complexity and uncertainty, and estimate again until the team reaches a shared value.

Story points represent relative effort, complexity and uncertainty, so they are not hours or days. A story receives a higher estimate when it involves more components, dependencies, technical uncertainty or testing. If a story is too large or unclear to estimate confidently, the team should clarify or split it before committing to it. Timeboxed research activities are recorded as spikes rather than ordinary user stories.

4.2 Reference Stories & Story Point Scale

The following existing backlog stories provide reference points for consistent estimation.

Points	Reference story	Reason
3	US-01: Register as a job seeker or employer	A relatively small, well-defined account-registration capability with a limited workflow.
5	US-03: Create and maintain a job seeker profile	Requires profile fields and handling of skills, qualifications, and experience.
8	US-19: Analyze a job seeker’s resume, qualifications, skills, and experience	A technically complex story involving resume/profile analysis and greater uncertainty.

Story points	Meaning
1	Very small and simple change
2	Small story with clear requirements
3	Medium story that needs some design and testing
5	Large story with several parts or connections to other features
8	Very large or complex story with some uncertainty

4.3 Team Capacity for the Sprint

The current team plan estimates capacity at approximately 20–25 story points per two week sprint. This is the initial planning capacity stated in the team draft. Because no completed sprint results are available yet it should not be presented as measured historical velocity.

The six stories currently assigned to Sprint 1 in the product backlog total 22 story points. This is within the stated 20–25-point capacity range, assuming the planned team availability and that the selected stories can meet the Definition of Done.

Sprint 1 story	Points
US-01**:** User registration	3
US-02**:** Secure login and logout	3
US-03**:** Job seeker profile	5
US-04**:** Employer profile	5
US-05**:** Role-based access permissions	3
US-06**:** Hosting within Saudi Arabia	3
Total planned Sprint 1 workload	22

4.4 Forecasted Velocity & How It Was Calculated

Since this is the team initial sprint actual velocity cannot yet be calculated from historical performance. The team will use its stated capacity range of 20–25 story points as an initial forecast. The midpoint is:

(20 + 25) ÷ 2 = 22.5 story points per sprint

Therefore, the initial forecast is approximately 22–23 story points per two-week sprint, with a planning range of 20–25 points. Sprint 1 planned 22 points fall within this

range. At the end of each sprint, actual velocity will be calculated by adding the points of stories fully completed according to the Definition of Done. Incomplete stories do not count as completed velocity and return to the Product Backlog for re-planning. The forecast should be updated using actual results from subsequent sprints.

5. Sprint 1

5.1 Sprint 1 Goal

Establish the secure foundation of the Fursah platform by enabling job seekers and employers to register and access their accounts, create their initial profiles, apply the correct role based permissions and ensure that the platform data and infrastructure are hosted within Saudi Arabia. At the end of the sprint, the team should have a working foundation that supports the development of job search and recruitment features in later sprints.

5.2 Sprint 1 Dates & Duration

Item	Sprint 1 plan
Duration	2 weeks
Schedule in current project timeline	Weeks 3–4, following Sprint 0 preparation
Calendar dates	Not specified in the current draft; insert the team-agreed dates if required.
Planned workload	22 story points
Estimated capacity	20–25 story points
Sprint Review	At the end of Week 4; demonstrate completed stories and collect stakeholder feedback.
Sprint Retrospective	After the Sprint Review; identify improvements for the next sprint.

5.3 Selected User Stories

The following stories are selected because they are already assigned to Sprint 1 in the team's prioritized product backlog. Their total estimate is 22 story points.

ID	User story	SP	Priority
US-01	As a user, I want to register as a job seeker or an employer, so that I can access Fursah's features.	3	High

US-02	As a registered user, I want to log in and out securely, so that I can access my account safely.	3	High
US-03	As a job seeker, I want to create and maintain a profile with my skills, qualifications, and experience, so that the platform can find opportunities that suit me.	5	High
US-04	As an employer, I want to create and maintain an employer profile, so that job seekers can learn about my organization.	5	High
US-05	As a platform administrator, I want job seekers, employers, and administrators to have different access permissions, so that each user only uses the features meant for them.	3	High
US-06	As the Ministry (HRSD), I want platform data and infrastructure hosted within Saudi Arabia, so that the project meets its hosting requirement.	3	High
	Total	22	

5.4 Acceptance Criteria

The following criteria define when each selected Sprint 1 story can be accepted. They elaborate the existing backlog stories without introducing later-release functionality.

US-01: User registration

- A visitor can choose to register as either a job seeker or an employer.
- The registration form requests the required account information and validates mandatory fields.
- Invalid or incomplete submissions are rejected with clear validation messages.
- A successful registration creates an account with the correct user role.
- The user receives confirmation that registration succeeded and can proceed to login.

US-02: Secure login and logout

- A registered user can log in using valid credentials.
- Invalid credentials are rejected and an appropriate error message is displayed without exposing sensitive information.
- After login, the user is authenticated and can access only functions permitted for their role.
- The user can log out, and the authenticated session is ended.

- Protected pages cannot be accessed through an ended or expired session without logging in again.

US-03: Job seeker profile

- An authenticated job seeker can create a profile.
- The profile allows the job seeker to enter and save skills, qualifications, and experience.
- Required fields are validated, with clear messages for missing or invalid information.
- The job seeker can view and update saved profile information.
- Profile information is associated with the correct account and is not accessible to unauthorized roles.

US-04: Employer profile

- An authenticated employer can create an employer profile.
- The profile allows the employer to enter and save the organization information required by the platform.
- Required fields are validated, with clear messages for missing or invalid information.
- The employer can view and update the saved organization profile.
- The profile is associated with the correct employer account and is not editable by unauthorized users.

US-05: Role-based access permissions

- The system distinguishes job seeker, employer, and administrator roles.
- Each role can access only pages and actions authorized for that role.
- An unauthorized attempt to access a restricted page or action is denied.
- Changing a page URL or sending a direct request does not bypass role permissions.
- Access control behavior is tested for all three roles.

US-06: Hosting within Saudi Arabia

- The deployment and hosting arrangement for platform data and infrastructure is documented and verified as located within Saudi Arabia.
- The selected hosting environment and relevant service-provider information are recorded for project review.
- The team verifies that planned storage and processing arrangements meet the Saudi hosting requirement.
- Evidence of the hosting location and configuration is available for review by authorized project stakeholders.

Definition of Done and Sprint Review

In addition to the story specific acceptance criteria, each story must satisfy the team definition of done: implementation is reviewed by another developer, all required tests pass, acceptance criteria are met, and the Product owner approves the completed work. For stories involving personal information, the implementation must also follow the project's Saudi data protection requirements. The Sprint Review demonstrates completed functionality to relevant stakeholders and records their feedback.

6. Summary of Changes to Previous Artifacts

6.1 Changes from Artifact 1

The table below summarizes how the baseline scope, assumptions, and phase definitions established in Artifact 1 were operationalized and refined in Artifact 3:

Project Area	Artifact 1 Baseline	Artifact 3 Operationalization
Scope Definition	Broad feature lists and high-level acceptance criteria for Job Seeker, Employer, and Matching modules.	Deconstructed into a 52-item Product Backlog (49 User Stories + 3 Spikes) totaling 236 Story Points.

Non-Functional Requirements	High-level assumptions regarding PDPL data privacy, Saudi hosting, and bilingual support.	Formatted into concrete, deliverable stories within the **Shared Platform Services** epic (US-05 to US-06, US-26 to US-30).
Project Phases & Roadmap	Static 3-phase rollout (*Core*, *Expansion*, *National Rollout*).	Restructured into Sprint 0 plus 11 two-week sprints across 3 releases: Release 1 (Sprints 1–6), Release 2 (Sprints 7–9), and Release 3 (Sprints 10–11).
Technical Risks	Identified high-risk features like NLP parsing and government service integrations as standard deliverables.	Introduced timeboxed Research Spikes (SP-01, SP-02, SP-03) to conduct technical investigations prior to development.
Work Decomposition	Broad deliverables without breakdown structure.	Formalized via a 7-phase Hybrid WBS covering planning, design, dev, testing, integration, and scope review.
Quality & Review	End-of-phase milestone acceptance.	Designated Sprint 6 with a lighter planned load (13 SPs, below the 20–25 SP capacity) dedicated to integration, PDPL security testing, and PO review.
Bilingual Support	Listed under the National Rollout phase, and as a platform-wide constraint.	Moved to Release 1 (Sprint 5, US-26), since adding it only at national rollout would require reworking features already delivered.

6.2 Changes to Artifact 2

Four types of development methodologies have been studied by the team in Artifact 2 to identify which methodology would be the best fit for Fursah. Two types of Plan-Driven Development Methodologies and two Agile Development Methodologies have been analyzed based on their advantages and disadvantages, compatibility with the project and organization, and related risks. Finally, Scrum Development Methodology has been chosen.

In Artifact 3, the Scrum approach selected in Artifact 2 was developed into a more detailed implementation and scheduling plan. The general decision to use Scrum remained

the same, but the team added practical details explaining how it would be applied throughout the project.

The main additions include:

- Product Backlog: The team came up with a prioritized product backlog which consisted of 49 user stories and 3 research spikes that were categorized into epics. Story points were used to estimate the work required per user story with a total of 236 story points.
- Sprint Planning: Two weeks were set aside for each sprint while the capacity of story points per sprint was estimated at 20-25 story points.
- Scrum Activities: The team identified the duration and purpose of the activities Sprint Planning, Daily Scrum, Sprint Review, Sprint Retrospective, and Backlog Refinement.
- Other Scrum Artifacts: Ideas like Sprint 0, epics, research spikes, and Definition of Done were used to assist in defining how the team will prepare for the project, conduct themselves in the project, research on technical unknowns and evaluate their work.
- Release Planning: The first release was scheduled to run from Sprint 0 to 6 Sprints of development, and took approximately 14 weeks. Releases 2 and 3 were sketched out but will be further developed as they approach.
- Details about Sprint 1: Six user stories were chosen and had the total value of 22 story points and details about acceptance criteria for each user story were added.

In conclusion, Artifact 3 advances the choice of methodology presented in Artifact 2 by transforming the Scrum methodology that has been chosen into a development process. This is achieved by outlining sprints, estimation methods, activities, and acceptance criteria without changing the choice of using Scrum for Fursah.